

Travis Brown

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OBJECTIVE

Seeking internship in Embedded/Firmware Engineering from May 2027 to August 2027.

EDUCATION

Rochester Institute of Technology

Rochester, NY

Bachelor of Science and Master of Science in Computer Science (Accelerated BS/MS Program)

Expected: December 2027

GPA: 3.95

SKILLS

Languages: C, C++, Python, Rust, Java, JavaScript, SQL, MIPS Assembly

Embedded: STM32 / ESP32 microcontrollers, ThreadX RTOS, CAN, SPI, I2C, UART, PWM, ADC, OpenOCD, Logic Analyzer

Tools/Environments: Git, GitHub, GitLab, Linux, macOS

WORK EXPERIENCE

QA Automation Intern

Lockheed Martin

May 2025 – August 2025

- Produced a Python-based automation tool that streamlined static code review, reducing QA review time by 75% (from 20 mins to 5 mins).
- Automated over 60% of manual software validation checks, significantly reducing QA workload and improving team efficiency.
- Collaborated with QA engineers to identify and resolve software issues, improving release reliability and reducing defects.
- Built and maintained CI/CD pipelines using GitLab CI, contributing to faster and more reliable testing cycles.
- Improved scalability and maintainability of Python automation scripts by modularizing code and implementing unit tests.

Digital Content Assistant

Spire Business Inc

June 2024 – April 2025

- Created SEO-optimized web pages with HTML/CSS per owner's specifications.
- Resolved common computer issues, such as slow performance, application errors, and login difficulties, ensuring minimal downtime.

PROJECTS

Real-Time Vehicle Control Unit (VCU)

RIT Electric Vehicle Team

October 2025 – Present

- Engineered embedded firmware in C++ for a custom-designed PCB serving as the central VCU for an electric motorcycle.
- Wrote firmware on an STM32 microcontroller within a ThreadX RTOS environment, focusing on real-time coordination, deterministic behavior, and time-critical outputs.
- Implemented CAN and CANopen communication between the VCU, motor controller, power subsystem, battery management, and more.
- Contributed to safety-critical system behavior through fault handling, startup validation, and robust inter-module communication.
- Coordinated with electrical and mechanical engineers to define system interfaces, communication protocols, and fault-handling behavior.

Handheld Chip-8 Emulator Console

Personal Project

June 2025 – December 2025

- Developed a Chip-8 emulator in Rust for a Raspberry Pi, with display, input, and sound using GPIO and SPI communication.
- Designed a 4x4 matrix keypad PCB in KiCad, including schematic and layout, and manually soldered key switches, caps, and diodes.
- Created a custom SPI display driver for an SSD1309 OLED screen, avoiding external graphics libraries for low-level control.

STM32 F4 Peripheral Support

RIT Electric Vehicle Team

December 2023 – November 2024

- Implemented driver support for PWM, ADC, SPI, and timers on the STM32F4 microcontroller family, working with the STM32 Hardware Abstraction Layer.
- Leveraged STM32's CubeMX and CubeIDE to streamline configuration and integration with EVT's hardware-agnostic API.
- Applied microcontroller datasheets and reference manuals to develop reliable, reusable peripheral drivers.